

Yifei Chen

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GPA: 4.0 / 4.0

EDUCATION:

University of California, San Diego

Sep 2024 - Present

- B.S. Data Science
- GPA: 4.0 / 4.0
- Honors: Marshall College Honors Program; Provost Honors (All Quarters to Date)

The Experimental High School Attached to Beijing Normal University

Sep 2021 - Jul 2024

The Branch of the High School Affiliated to Renmin University of China

Sep 2018 - Jul 2021

WORK EXPERIENCE

HDSI Lab Fellow, UC San Diego

Nov 2025 – Present | ~ 10 hrs/week

- Developed embedded hardware prototypes using Arduino and Raspberry Pi integrating sensors for research experiments.
- Designed and fabricated custom mechanical components using CAD and 3D printing for experimental systems.
- Implemented microcontroller programming, circuit setup, debugging, and system testing to ensure reliable data capture.
- Key challenge: maintaining system stability under changing experimental requirements. Solved with modular hardware design and iterative validation.

Internship, Zhongdiantongtu Technology.,LTD, Wuhan, China

Jan 2023 – Feb 2023 | Full-time

- Worked under CTO Lei Cheng in the digital twin technology team for transportation systems.
- Assisted with data collection, simulation modeling, and traffic behavior prediction.
- Used EarthMaker to simulate tunnel traffic flow, signage placement, and air circulation.
- Conducted literature reviews of 10+ academic papers on digital twins and transportation modeling.
- Key challenge: mapping real-world assumptions into simulation parameters. Solved with iterative tuning and cross-validation with literature.

PROJECT EXPERIENCE

Geisel Library Occupancy Tracker — DS3 Project

Team Leader | Team of 4 | Sep 2025 – Dec 2025 | ~ 12 hrs/week

Skills: ESP32, Arduino, Embedded Systems, Full-Stack Development, React, Firebase, Real-time Systems

- Led the entire project lifecycle, participating in every technical component from hardware to frontend deployment.
- Designed and deployed ESP32-based Bluetooth scanning hardware, including Arduino firmware development, sensor configuration, and debugging.
- Built the full-stack web application using React and Firebase Realtime Database, deployed on Vercel, enabling real-time occupancy visualization.
- Designed the database schema, real-time listeners, and floor-plan-based visualizations.
- Coordinated team task allocation, integration, and testing as project lead.
- Challenges: 1. Unstable Bluetooth signals and noisy scans. Solved by implementing filtering heuristics, scan aggregation, and schema redesign. 2. Designing a visualization that matched Geisel Library's geometry. Solved by constructing the building with stacked bar-chart floor elements that fill inward in real time as occupancy increases.
- Demonstrated a complete physical AI pipeline, from sensing to embedded control to data infrastructure to real-time visualization.
- Won Second Place Project in Dino Cage (DS3 Project Competition)

- Website: <https://ds-3-geisel-library-project.vercel.app/>

Multimedia Processing and Classification System

Individual Project | Jan 2025 – Mar 2025 | ~ 8 hrs/week

Skills: *PyTorch, CNN, Deep Learning, K-Nearest Neighbors (KNN), Batch Training & Inference*

- Designed an object-oriented multimedia processing system supporting image transformations and audio effects with tiered usage constraints.
- Implemented an end-to-end image classification pipeline in PyTorch, including custom Dataset/DataLoader, tensor preprocessing, CNN modeling, training, and batch-based inference.
- Built and trained a convolutional neural network (CNN) for image classification, alongside a K-Nearest Neighbors (KNN) baseline to compare traditional and deep learning approaches.
- Utilized batch processing to improve training efficiency and inference throughput.
- Implemented model checkpointing to save and reload trained networks for reproducible evaluation.
- Challenges: 1. Balancing model accuracy with computational efficiency. Solved by lightweight CNN design, batch inference, and KNN baseline comparison. 2. Efficient multimedia classification under compute constraints. Solved by batch processing and lightweight CNN architectures.

Music Popularity Analysis

Individual Project | Mar 2025 – Jun 2025 | ~ 8 hrs/week

Skills: *PyTorch, Neural Network Regression, LLM API Integration (Gemini)*

- Analyzed large-scale TikTok and Spotify music datasets to identify factors influencing song popularity.
- Performed data cleaning, feature engineering, standardization, and train/test evaluation using Python, Pandas, and NumPy.
- Built and trained PyTorch-based neural network regression models to predict track popularity from audio features and artist metrics.
- Evaluated model performance using quantitative metrics and structured error analysis.
- Integrated the Gemini API to generate automated analytical reports from structured model outputs, correlations, and prediction errors.
- Designed prompts that combined numerical model results with metadata to produce interpretable, human-readable insights.
- Challenge: 1. Translating high-dimensional model outputs into meaningful explanations. Solved by coupling PyTorch predictions with LLM-generated analytical summaries. 2. Model interpretability for complex popularity features. Solved by translating predictions into structured LLM-based insights.

Taylor Swift Music Data Analysis

Individual Project | Sep 2024 – Oct 2024 | ~ 6 hrs/week

Skills: *Text Analysis, Data Visualization, Feature Extraction*

- Analyzed lyrics from 200+ songs to identify thematic and popularity trends.
- Built a lyric-based song recommendation tool using similarity metrics and keyword search.
- Created data visualizations and keyword clouds for album-level analysis.
- Key challenge: extracting meaningful structure from unstructured text. Solved by preprocessing and feature extraction.

Friends TV Show Data Analysis

Individual Project | Oct 2024 – Dec 2024 | ~ 6 hrs/week

Skills: *Sentiment Analysis, Statistical Analysis, Data Processing*

- Analyzed episode-level datasets to study dialogue gender balance, character emotions, and viewership trends.
- Performed sentiment analysis and statistical correlation analysis.
- Generated emotion distributions and algorithmically created new episode titles.
- Key challenge: noisy natural-language data. Solved by preprocessing pipelines and aggregation strategies.

Mechanical Subteam Member, TritonBots RoboCup SSL Team

Team Member | Team of 6 | Sep 2025 – Present | 5 ~ 7 hrs/week

Skills: *Mechanical Design, CAD, Robotics Systems Integration*

- Designed and prototyped robotic mechanisms like the dribbler damping system of competition robots.
- Conducted CAD modeling, material selection, and iterative refinement based on testing feedback.
- Collaborated with electrical and software subteams to integrate mechanical components into the full assembly of the robot.

- Key challenge: maintaining stability under dynamic motion. Solved by damping optimization and rapid prototyping.

International Space Settlement Design Competition

Mechanical Designer & 3D Modeling Lead | Oct 2021 – Aug 2023

Skills: Mechanical Design, Large-Scale 3D Modeling, Cross-Team Collaboration

- Led the 3D modeling in three out of four departments (Human Factor Department, Automation Department, and Operation Department).
- Helped the teams from different countries communicate on window composition, radiation protection, construction machinery, commodities & provision, utilities & transportation routings, and energy generation & storage.
- Presented the project to the ISSDC committee; won the International Finals.
- 3D printed the main model, set up a project booth and presented the project at the Award Ceremony of 2023 UNESCO (United Nations Educational Scientific and Cultural Organization) Prize for Girls' and Women's Education.

Mechanical Engineer of VEX Robotics Team

Team Member | Sep 2021 – Jul 2023 | 4 hrs/week

Skills: Mechanical Design, CAD, Robotics Prototyping

- Designed, prototyped, and assembled mechanical structures like drivetrain, intake, and lift mechanisms for VEX robots.
- Worked with another mechanical engineer of the team to develop efficient and reliable designs under strict time and performance constraints.
- Used CAD tools for 3D modeling and part optimization, also integrated feedback from testing to improve robot performance.
- Contributed to team's overall strategy and documentation for regional competitions to ensure mechanical consistency and team coordination.

RESEARCH EXPERIENCE

Research on x-y plotter

Researcher | Oct 2022 – Sep 2023 | 6 hrs/week

Skills: Motion Control, Arduino, Servo & Stepper Motors, Mechatronic System Design

Adviser: Prof. Michael Littman, Princeton University

- Researched and designed an x-y plotter system using servo and stepper motors.
- Modeled designs using LeoCAD and built prototypes with LEGO Technic components.
- Programmed motion control using Arduino IDE and successfully plotted data and text.
- Reviewed 15+ technical references on CNC systems and control theory.
- Authored a 15-page research paper documenting design, implementation, and testing.

AWARDS & HONORS

- Provost Honors, UC San Diego — All Quarters to Date
- Marshall College Honors Program (UC cumulative GPA ≥ 3.8)
- Winner, International Space Settlement Design Competition (ISSDC)
- Second Place, Dino Cage Project Competition (DS3)
- Gold Medal, iGEM (International Genetically Engineered Machine Competition)
- Honor Roll of Distinction, AMC 12 A & B; USAMO Cutoff (Top 1% Worldwide)

LEADERSHIP

Founder, Bibliolater Club, Beijing

Founder | Sep 2022 – Jun 2024

- Founded and led a reading club with 40+ members from multiple elite high schools.
- Organized 120+ online meetings and facilitated discussions across 40 books.
- Defined club mission focused on collaborative reading and idea exchange.

SKILLS

Languages: Mandarin, English, Spanish

Programming Languages & AI: Python, PyTorch (CNNs, neural network regression, training & inference pipelines), Java, JavaScript, MATLAB, HTML

Machine Learning & Data Science: Tensor preprocessing, batch-based training & inference, feature engineering, model evaluation (MSE, MAE, R^2), error analysis, correlation analysis, train–test splitting

AI Systems & APIs: Gemini API (LLM integration, prompt design, automated report generation from structured ML outputs)

Web & Data Technologies: Pandas, NumPy, Matplotlib, React, Firebase, JSON, CSV

Embedded Systems & Hardware: Arduino, ESP32, Raspberry Pi, sensor integration, hardware–software co-design

Engineering & Design: SolidWorks, CAD, Blender, SketchUp, 3ds Max, Cinema 4D

Fabrication: 3D Printing, Carpentry, Soldering, Laser Cutting

Software & Development Tools: Git, Git Bash, PyCharm, Jupyter Notebook, virtual environments (venv)

Productivity Tools: Microsoft Word, Excel, PowerPoint